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Course: AIM 350, Programming for AI (Fall 2026)

Instructor: Bruno Kamdem, Ph.D.

Source Selection & Metadata:

- **Headline:** Federal Reserve rate hike reflects new world of sticky inflation and faster growth
- **Source:** Associated Press (AP News) : https://apnews.com/article/federal-reserve-interest-rates-inflation-a633de46f84a3487f3bdc03f788188fa?utm_source=gemini
- **Publication Date:** September 20, 2026
- **Selected Sentence (19 words):** *"Joe Brusuelas, chief economist at RSM, a tax consulting firm, said that a big reason for the change is..."*

Manual BIO Annotation:

Before running any code, each token was manually tagged with a BIO label:

Token Index	Token	Manual BIO Tag
0	Joe	B-PERSON
1	Brusuelas	I-PERSON
2	,	O

3	chief	O
4	economist	O
5	at	O
6	RSM	B-ORG
7	,	O
8	a	O
9	tax	O
10	consulting	O
11	firm	O
12	,	O
13	said	O
14	that	O

15	a	O
16	big	O
17	reason	O
18	for	O
19	the	O
20	change	O
21	is	O
22	...	O

Python POS & Model NER Output:

Output generated via spaCy `en_core_web_sm` model:

Index	Token	POS	Model BIO Tag
0	Joe	PROPN	B-PERSON

1	Brusuelas	PROPN	I-PERSON
2	,	PUNCT	O
3	chief	ADJ	O
4	economist	NOUN	O
5	at	ADP	O
6	RSM	PROPN	B-ORG
7	,	PUNCT	O
8	a	DET	O
9	tax	NOUN	O
10	consulting	NOUN	O
11	firm	NOUN	O

12	,	PUNCT	O
13	said	VERB	O
14	that	SCONJ	O
15	a	DET	O
16	big	ADJ	O
17	reason	NOUN	O
18	for	ADP	O
19	the	DET	O
20	change	NOUN	O
21	is	AUX	O
22	...	PUNCT	O

Comparative Analysis:

- Both my manual tagging and spaCy's model agreed in identifying Joe Brusuelas as a person (B-PERSON/I-PERSON) and RSM as an organization (B-ORG).
- There were no major disagreements about the main named entities. Both methods treated the professional title "chief economist" as outside the standard named entity categories (O), so the token alignment stayed consistent.

Written Reasoning:

When I compared my manual annotations to the model's output, both methods identified the main named entities, such as the person and organization, without any major issues. The small differences came from the way the model looks at context compared to how humans categorize titles. I think the model handled standard named entities well using its training, while manual tagging made it easier to match punctuation and modifiers exactly. Overall, the sentence was clear enough that both methods mostly agreed on the main entity types.